PROBLEM DEFINITION

Markets are a network.
No one is treating them as one.

Institutional investors spend $400K–$900K per year per desk on Bloomberg, FactSet, and AlphaSense — and still cannot answer the most important question in finance: when something happens to Company X, which companies are affected, by how much, and why?

WHAT EXISTS TODAY

Individual Entity Intelligence

Bloomberg, FactSet, LSEG, and AlphaSense provide deep real-time data, news, and analytics on individual companies — prices, fundamentals, sentiment, filings, earnings transcripts.

WHAT IS MISSING

Relationship Intelligence

No commercial platform models how companies are connected, how those connections change over time, or how a shock to one entity propagates through its network of dependencies to affect others.

THE COST OF THE GAP

Delayed Second-Order Reactions

Second and third-order effects — supply chain contagion, leadership network exposure, macro sensitivity chains — are discovered manually, hours or days after the market has already moved.

CORE PROBLEM STATEMENT

The market prices individual entity news quickly. It prices relationship-driven second and third-order effects slowly. That lag is the alpha opportunity — and it is invisible without a relationship graph.

Capital IQ has a static executive network. AlphaSense has documents. Bloomberg has prices. None of them have a dynamic, weighted, queryable graph of 5,000+ global market entities with a neural network that adjusts relationship strengths in real time from incoming news signals.

THE PROPAGATION PROBLEM — HOW A SHOCK TRAVELS THROUGH THE NETWORK

Known edge Correlation edge Blind spot (no current system shows this) -73 Impact score (0–100)

THE FOUR CORE PROBLEMS

What the market fails to provide

01 STATIC RELATIONSHIPS

Every existing system treats company relationships as binary and unchanging. A supply chain link between TSMC and NVIDIA exists or it doesn't. But in reality, that link's strength changes every time TSMC reports earnings, every time geopolitical tension rises, every time a new fab is announced.

NO DYNAMIC WEIGHTS

02 NO PROPAGATION ENGINE

Capital IQ shows you that Person A sits on Board B. But it doesn't tell you: given that news event N just hit Company C, traverse the graph and score every entity that is reachable within 3 hops, weighted by edge strength and time decay.

NO SIGNAL PROPAGATION

03 NO TEMPORAL MEMORY

No platform lets you ask: "What did the relationship graph look like the day before SVB collapsed? Which edges were strengthening? Which were anomalously quiet?" Without temporal graph state, backtesting is impossible to do correctly.

NO TIME-TRAVEL

04 BLACK-BOX OR MANUAL

Quant signals exist (momentum, factor models) but have no explainability. Human research exists but cannot scale to 5,000+ companies globally. There is no middle ground: a scalable, automated system that still shows its full reasoning chain per output.

NO EXPLAINABILITY

✓ — WHAT EXISTING SYSTEMS DO WELL

- ✅ Real-time price and volume data (Bloomberg, LSEG)
- ✅ Deep fundamental company data (FactSet, Capital IQ)
- ✅ AI document search across millions of filings (AlphaSense)
- ✅ Static executive and board network mapping (Capital IQ)
- ✅ Traditional quant factor signals (momentum, value, quality)
- ✅ Portfolio risk models (Barra, Axioma via FactSet)

✗ — WHAT NO EXISTING SYSTEM DOES

- ❌ Dynamic edge weights adjusted by neural network from live news
- ❌ Automated event propagation: impact score + direction + time decay
- ❌ Full temporal graph rewind to any past timestamp
- ❌ On-premise, air-gapped deployment with data sovereignty
- ❌ Full explainability chain: signal → edges → news → source doc
- ❌ Autonomous multilingual scraping agent (EN/ZH/JA/DE/FR)

THE ALPHA OPPORTUNITY

The relationship-driven second and third-order effects of news events are consistently underpriced by the market — because no systematic tool exists to score and surface them at institutional speed. That is the gap MKG fills.

$35B
Spent annually on financial data — without this layer

5,000+
Companies in production universe

7 gaps
No competitor fills more than 2–3 simultaneously

< 60s
Target latency from news to signal delivery

SEVEN CRITICAL GAPS

What MKG is built to fill

01
Dynamic Neural-Network-Adjusted Edge Weights
Edge weights update in real time from news signals. TSMC–NVIDIA supply edge strengthens the moment TSMC misses earnings. No existing system does this.
CORE IP

02
Event Impact Propagation with Quantified Output
Impact score (0–100) + direction (±) + confidence + time decay curve per affected company. Automated graph traversal, not manual research.
CORE IP

03
Full Temporal Graph State History & Rewind
Inspect graph at any past timestamp. Diff between T1 and T2. Replay any historical event frame by frame. Critical for valid backtesting.
DATA MOAT

04
On-Premise, Air-Gapped Deployment
Bloomberg, LSEG, AlphaSense are cloud-only. For data sovereignty, this disqualifies them. MKG runs entirely within the institution's own infrastructure.
SOVT. MOAT

05
Full Signal Explainability Chain
Every signal traces back: output → graph edges → weight delta → news event → source document. Compliant with institutional fiduciary and regulatory review.
REG. MOAT

06
Autonomous Multilingual Scraping Agent
Continuously scrapes global news in EN, ZH, JA, DE, FR, ES. No dependency on a single news vendor. Builds a proprietary corpus over time that competitors cannot replicate.
DATA MOAT

07
Purpose-Built for Alpha Generation — Not Compliance, Not Search
The only system designed from the ground up with alpha generation as its north star. Every design decision — the output schema, the backtest gate, the multi-horizon architecture, the portfolio layer — exists to generate returns, not manage data.
NORTH STAR

THE MKG SOLUTION — ARCHITECTURE PREVIEW

Four layers that solve the problem end-to-end

LAYER 01
Intelligence Agent
Hybrid scraping agent. Continuous for breaking news, scheduled for regional press. Multilingual NLP pipeline. Entity extraction → event classification → impact hypothesis.

LAYER 02
Knowledge Graph
Versioned heterogeneous graph. 6 node types, 7 edge types, multi-relational. Two-speed versioning: slow structural + fast weight time-series with full history.

LAYER 03
Weight Adjustment Network
GAT-based neural network. Trained on historical events first. Adjusts edge weights from incoming signals. Multi-horizon output: 1h / 1d / 1w / macro. Backtest-gated before live.

LAYER 04
Output & Debug Layer
Dashboard + push alerts + REST/WebSocket API. Full signal explainability. Temporal replay UI. Model health monitor. Complete audit trail per signal.